Tsz Fei Wang

Kitchener, Waterloo | wangtszfei@gmail.com | 647-939-3797 | LinkedIn | GitHub

Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages: Python, Java, C, C++, TypeScript, JavaScript, SQL
Frameworks & Tools: React, Next.js, Node.js, FastAPI, PyTorch, Pandas, Git, Linux, Docker, GitHub Actions

Projects

huddle | *Node.js, Cloudflare Workers, MCP* — *Oct 2026 – Present*

- Built a tool that lets several **Claude Code** agents on different machines work in one repo at once. Added **per-file locks**, so while one agent edits a file the other copies go read-only and get its edits streamed in live.
- Wrote the relay server in plain **Node.js** with no dependencies and deployed it on **Cloudflare Workers**, then added agent-to-agent chat, a web dashboard and git history sync between checkouts.

Physical Studio | *Python, React, Three.js, Docker* — *Winner, Hack the Valley 11 – Oct 2026*

- Wrote the **MCP server** that lets an LLM design a hardware project from a library of measured parts and check its own design for collisions before exporting an animated **3D assembly guide**.
- Deployed the app with **Docker Compose** and Caddy on a Vultr server at physicalstudio.work. Team of three.

Technova Hackathon Badge | *KiCad, Hardware Lead* — *Oct 2026*

- Led the hardware design in **KiCad** of an NFC event badge for about **300 attendees** (RISC-V MCU, LED ring, USB-C charging), and routed the double-sided board to a clean DRC.
- Put together the **JLCPCB** order (gerbers, BOM matched to LCSC parts, placement file) and priced each option against the per-badge budget.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Wrote the **GLSL** shaders, a small water simulation and a live system HUD for an animated koi theme for KDE Plasma (**~1,290 downloads**). Fixed blank rendering on NVIDIA/Wayland after a user report by porting the shaders to GLSL ES.

Bing Bang Dream Adventure | *Godot, GDScript* — *2nd Place, Horizons Polaris – Aug 2026*

- Wrote the game engine, beatmap system and hit-timing logic for a rhythm game played with custom **IMU-based flick controllers**, matching each note to the sensor input in real time.

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Designed and implemented a **point-in-time backtesting pipeline** (**Python**, **pandas**, yfinance), eliminating lookahead bias in **LLM-driven** equity research.
- Built a **model-agnostic evaluation framework** benchmarking stock-preference and inter-stock-relationship predictions across **9 LLM architectures** and **104 tickers**, isolating true signal quality from raw model capability.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built a **full-stack web data access portal** for Catrin (Catalogue of Transcriptional Regulatory Interactions), unifying **~4 million** TF-TG associations across **33,000 genes** from **9** independent data sources into a single queryable interface.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Designed and delivered **cybersecurity** curriculum and class presentations reaching **60+ students** to raise digital-safety awareness school-wide.
- **Captained team to a national #1 finish** in CyberTitan, Canada's largest **cyberdefense** competition, coordinating strategy and technical training across the roster.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*

- Built an **audio deepfake detection** model in **Python** on **Kaggle**, training and evaluating a classifier that distinguishes real from AI-synthesized speech.

Off Script – Waterloo Tech Week | *Founder & Host, 120+ registrations* — Sept 2026

Awards & Honors

Gold & Silver Medalist, IT Systems Networking – Skills Ontario (Regional & Provincial) — Mar – May 2026
National Champion x2 – 1st in Canada, Cybersecurity – $1,000 scholarship — May 2025, Mar 2026
National Finalist – 2nd in Canada, Cybersecurity – $1,600 scholarship — Mar 2025
1st Place, Canada, Mirai Project – Technology Innovation in Healthcare – $400 award — Jan 2025